1 Truth

($c_\perp$ published wall curve; env $= [\bar{w}_{lo},\ \bar{w}_{hi}]$ planned envelope; $\bar{w}_0$ start height; w_c contact height)

$$\bar{a}_{\text{true}}(\bar{w}) = \frac{1}{c_\perp(\bar{w})}$$
$$b_{\text{true}}(\bar{w}) := \frac{\bar{a}'_{\text{true}}(\bar{w})}{\big(1-\bar{a}_{\text{true}}(\bar{w})\big)^2}$$
$$\frac{1}{1-\bar{a}_{\text{true}}(\bar{w})} = \frac{1}{1-\bar{a}_{\text{true}}(\bar{w}_0)} + \int_{\bar{w}_0}^{\bar{w}} b_{\text{true}}(s)\,ds$$
$$b'_{\text{true}}(\bar{w}) := \frac{db_{\text{true}}}{d\bar{w}}$$

$$c_\perp(\bar{w}) = 1 + \frac{9}{8\,\bar{w}} + \mathcal{O}\big(\bar{w}^{-2}\big)$$
$$\lim_{\bar{w}\to\infty} b_{\text{true}}(\bar{w}) = \frac{8}{9}, \qquad b_{\text{true}}(\bar{w}) = \frac{8}{9} + \mathcal{O}\big(\bar{w}^{-2}\big), \qquad b'_{\text{true}}(\bar{w}) = \mathcal{O}\big(\bar{w}^{-3}\big)$$
$$\bar{a}_{\text{true}}(w_c) = 0, \qquad \bar{a}'_{\text{true}}(w_c) = 1, \qquad b_{\text{true}}(w_c) = 1$$
$$\sup_{\text{env}} \big|b'_{\text{true}}\big| = \big|b'_{\text{true}}(\bar{w}_{lo})\big|$$

$$b'_{\text{true}}(\bar{w}) \ \to\ b_1$$
$$b_{\text{true}}(\bar{w}) \ \approx\ b(\bar{w}_0) + b_1\,(\bar{w}-\bar{w}_0)$$
$$b_1\big|_{\bar{w}\to\infty} = 0, \qquad b_1\big|_{\bar{w}=\bar{w}_{lo}} = b'_{\text{true}}(\bar{w}_{lo})$$

2 Gain law

(bars denote normalized quantities: lengths by R, gain by a_{nom}, forces by a_o, $\bar{a}\,\bar{f} = \Delta\bar{w}$; hats on estimates; $e_\theta = \theta - \hat{\theta}$)

$$\bar{a}'(\bar{a}, b) := \frac{d\bar{a}}{d\bar{w}} = b\big(1-\bar{a}\big)^2$$
$$b_1 := \frac{db}{d\bar{w}}$$
$$\frac{db_1}{d\bar{w}} = 0$$
$$\bar{a}''(\bar{a}, b, b_1) := \frac{d\bar{a}'}{d\bar{w}} = b_1\big(1-\bar{a}\big)^2 - 2\,b^2\big(1-\bar{a}\big)^3$$

$$\frac{d}{d\bar{w}}\,\frac{1}{1-\bar{a}} = b$$
$$b(\bar{w}) = b(\bar{w}_0) + b_1\,(\bar{w}-\bar{w}_0)$$
$$\frac{1}{1-\bar{a}(\bar{w})} = \frac{1}{1-\bar{a}(\bar{w}_0)} + b(\bar{w}_0)\,(\bar{w}-\bar{w}_0) + \tfrac{1}{2}\,b_1\,(\bar{w}-\bar{w}_0)^2$$

3 True dynamics

$\big(\delta\bar{w}_j[k] := \delta\bar{w}[k-d+j-1],\ j=1,\dots,d+1,\ d=2\big)$

$$
\begin{aligned}
\bar{w}[k] &= \bar{w}_d[k] - \delta\bar{w}_3[k] \\
\bar{w}_d[k+1] &= \bar{w}_d[k] + \Delta\bar{w}_d[k] \\
\delta\bar{w}_1[k+1] &= \delta\bar{w}_2[k] \\
\delta\bar{w}_2[k+1] &= \delta\bar{w}_3[k] \\
\delta\bar{w}_3[k+1] &= \lambda_c\,\delta\bar{w}_3[k] - F_{dw}[k]\,e_{\bar{a}_w}[k] - \varepsilon_w[k] \\
\Delta\bar{w}[k] &= \Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\bar{w}_3[k] + F_{dw}[k]\,e_{\bar{a}_w}[k] + \varepsilon_w[k] \\
\bar{a}_w[k+1] &= \bar{a}_w[k] + b[k]\big(1-\bar{a}_w[k]\big)^2\,\Delta\bar{w}[k] \\
b[k+1] &= b[k] + b_1[k]\,\Delta\bar{w}[k] \\
b_1[k+1] &= b_1[k]
\end{aligned}
$$

$$
F_{dw}[k] = \bar{f}_{dw}[k] + (1-\lambda_c)\sum_{i=1}^{d}\bar{f}_{dw}[k-i]
$$

$$
\varepsilon_w[k] = \bar{a}_w[k]\,\bar{f}_T[k] + (1-\lambda_c)\sum_{i=1}^{d}\bar{a}_w[k-i]\,\bar{f}_T[k-i] + (1-\lambda_c)\,n_w[k]
$$

4 Estimator

$$
\mathbf{x} = \big[\,\delta\bar{w}_1,\ \delta\bar{w}_2,\ \delta\bar{w}_3,\ \bar{a}_w,\ b,\ b_1\,\big]^\top \qquad (d=2)
$$

$$
\begin{aligned}
\delta\hat{\bar{w}}_1[k+1] &= \delta\hat{\bar{w}}_2[k] \\
\delta\hat{\bar{w}}_2[k+1] &= \delta\hat{\bar{w}}_3[k] \\
\delta\hat{\bar{w}}_3[k+1] &= \lambda_c\,\delta\hat{\bar{w}}_3[k] \\
\Delta\hat{\bar{w}}[k] &= \Delta\bar{w}_d[k] + (1-\lambda_c)\,\delta\hat{\bar{w}}_3[k] \\
\hat{\bar{a}}_w[k+1] &= \hat{\bar{a}}_w[k] + \hat{b}[k]\big(1-\hat{\bar{a}}_w[k]\big)^2\,\Delta\hat{\bar{w}}[k] \\
\hat{b}[k+1] &= \hat{b}[k] + \hat{b}_1[k]\,\Delta\hat{\bar{w}}[k] \\
\hat{b}_1[k+1] &= \hat{b}_1[k]
\end{aligned}
$$

5 Error dynamics

(True − Estimator; coefficients at the estimate)

$$
e_{\Delta\bar{w}}[k] = \Delta\bar{w}[k] - \Delta\hat{\bar{w}}[k] = (1-\lambda_c)\,e_{\delta\bar{w}_3}[k] + F_{dw}[k]\,e_{\bar{a}_w}[k] + \varepsilon_w[k]
$$

$$
\begin{aligned}
b\big(1-\bar{a}_w\big)^2 &= \hat{b}\big(1-\hat{\bar{a}}_w\big)^2 - 2\,\hat{b}\big(1-\hat{\bar{a}}_w\big)\,e_{\bar{a}_w} + \big(1-\hat{\bar{a}}_w\big)^2 e_b + \mathcal{O}\big(e^2\big)\\
b\big(1-\bar{a}_w\big)^2\,\Delta\bar{w} &= \hat{b}\big(1-\hat{\bar{a}}_w\big)^2\,\Delta\hat{\bar{w}} + \hat{b}\big(1-\hat{\bar{a}}_w\big)^2\,e_{\Delta\bar{w}} - 2\,\hat{b}\big(1-\hat{\bar{a}}_w\big)\,\Delta\hat{\bar{w}}\,e_{\bar{a}_w} + \big(1-\hat{\bar{a}}_w\big)^2\,\Delta\hat{\bar{w}}\,e_b + \mathcal{O}\big(e^2\big)\\
b_1\,\Delta\bar{w} &= \hat{b}_1\,\Delta\hat{\bar{w}} + \hat{b}_1\,e_{\Delta\bar{w}} + \Delta\hat{\bar{w}}\,e_{b_1} + \mathcal{O}\big(e^2\big)
\end{aligned}
$$

$$
\begin{aligned}
e_{\delta\bar{w}_1}[k+1] &= e_{\delta\bar{w}_2}[k]\\
e_{\delta\bar{w}_2}[k+1] &= e_{\delta\bar{w}_3}[k]\\
e_{\delta\bar{w}_3}[k+1] &= \lambda_c\,e_{\delta\bar{w}_3}[k] - F_{dw}[k]\,e_{\bar{a}_w}[k] - \varepsilon_w[k]\\
e_{\bar{a}_w}[k+1] &= (1-\lambda_c)\,\hat{b}\big(1-\hat{\bar{a}}_w\big)^2 e_{\delta\bar{w}_3}[k] + \Big[1+\hat{b}\big(1-\hat{\bar{a}}_w\big)^2 F_{dw} - 2\,\hat{b}\big(1-\hat{\bar{a}}_w\big)\Delta\hat{\bar{w}}\Big]e_{\bar{a}_w}[k]\\
&\quad + \big(1-\hat{\bar{a}}_w\big)^2\Delta\hat{\bar{w}}\;e_b[k] + \hat{b}\big(1-\hat{\bar{a}}_w\big)^2\varepsilon_w[k]\\
e_b[k+1] &= (1-\lambda_c)\,\hat{b}_1\;e_{\delta\bar{w}_3}[k] + \hat{b}_1\,F_{dw}\;e_{\bar{a}_w}[k] + e_b[k] + \Delta\hat{\bar{w}}\;e_{b_1}[k] + \hat{b}_1\;\varepsilon_w[k]\\
e_{b_1}[k+1] &= e_{b_1}[k]
\end{aligned}
$$

$$
F_e[k] = \begin{bmatrix}
0 & 1 & 0 & 0 & 0 & 0\\
0 & 0 & 1 & 0 & 0 & 0\\
0 & 0 & \lambda_c & -F_{dw} & 0 & 0\\
0 & 0 & (1-\lambda_c)\,\hat{b}\big(1-\hat{\bar{a}}_w\big)^2 & \begin{array}{c}1+\hat{b}\big(1-\hat{\bar{a}}_w\big)^2 F_{dw}\\ -2\,\hat{b}\big(1-\hat{\bar{a}}_w\big)\Delta\hat{\bar{w}}\end{array} & \big(1-\hat{\bar{a}}_w\big)^2\Delta\hat{\bar{w}} & 0\\
0 & 0 & (1-\lambda_c)\,\hat{b}_1 & \hat{b}_1\,F_{dw} & 1 & \Delta\hat{\bar{w}}\\
0 & 0 & 0 & 0 & 0 & 1
\end{bmatrix}
$$

$$
\mathbf{q}[k] = \Big[\,0,\;0,\;-\varepsilon_w,\;\hat{b}\big(1-\hat{\bar{a}}_w\big)^2\varepsilon_w,\;\hat{b}_1\,\varepsilon_w,\;0\,\Big]^\top
$$

6 Measurement

$\big(\nabla_d\bar{w}_d[k] := \bar{w}_d[k] - \bar{w}_d[k-d]\big)$

$$
\begin{aligned}
y_1[k] &= \delta\bar{w}_1[k] + n_w[k] \\
y_2[k] &= \bar{a}_w[k-d] + n_a[k] \\
\bar{a}_w[k-d] &= \bar{a}_w[k] - b\big(1-\bar{a}_w[k]\big)^2\nabla_d\bar{w}_d[k] \\
&\quad + \tfrac{1}{2}\Big[b_1\big(1-\bar{a}_w[k]\big)^2 - 2\,b^2\big(1-\bar{a}_w[k]\big)^3\Big]\nabla_d\bar{w}_d[k]^2 + \mathcal{O}\big(\nabla_d\bar{w}_d^3\big) \\
\hat{y}_1[k] &= \delta\hat{\bar{w}}_1[k] \\
\hat{y}_2[k] &= \hat{\bar{a}}_w[k] - \hat{b}[k]\big(1-\hat{\bar{a}}_w[k]\big)^2\,\nabla_d\bar{w}_d[k] \\
H_{24}[k] &= \frac{\partial\hat{y}_2[k]}{\partial\hat{\bar{a}}_w[k]} = 1 + 2\,\hat{b}[k]\big(1-\hat{\bar{a}}_w[k]\big)\,\nabla_d\bar{w}_d[k] \\
H_{25}[k] &= \frac{\partial\hat{y}_2[k]}{\partial\hat{b}[k]} = -\big(1-\hat{\bar{a}}_w[k]\big)^2\,\nabla_d\bar{w}_d[k] \\
H_{26}[k] &= \frac{\partial\hat{y}_2[k]}{\partial\hat{b}_1[k]} = 0 \\
H[k] &= \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & H_{24}[k] & H_{25}[k] & H_{26}[k] \end{bmatrix} \\
\mathbf{e}_y[k] &= H[k]\,\mathbf{e}^-[k] + \begin{bmatrix} n_w[k] \\ n_a[k] \end{bmatrix}
\end{aligned}
$$

7 Noise

$$
\begin{aligned}
Q[k] &= Q_{33}[k]\,\mathbf{g}[k]\,\mathbf{g}[k]^\top, \qquad \mathbf{g}[k] = \Big[\,0,\ 0,\ -1,\ \hat{b}[k]\big(1-\hat{\bar{a}}_w[k]\big)^2,\ \hat{b}_1[k],\ 0\,\Big]^\top \\
Q_{55}[k] &= \hat{b}_1[k]^2\,Q_{33}[k], \qquad Q_{66} = 0 \\
Q_{33}[k] &= \kappa_T\Big(\hat{\bar{a}}_w[k] + (1-\lambda_c)^2\big(\hat{\bar{a}}_w[k-1] + \hat{\bar{a}}_w[k-2]\big)\Big) + (1-\lambda_c)^2\sigma_{n_w}^2, \qquad \kappa_T = \frac{4k_BT\,a_o}{R} \\
R[k] &= \mathrm{diag}\big(R_1,\ R_2[k]\big), \qquad R_1 = \sigma_{n_w}^2 = \frac{\sigma_n^2}{R^2}, \qquad R_2[k] = 2a_{\mathrm{cov}}^2\,IF_{\mathrm{var}}\big(\hat{\bar{a}}_w[k]\big)\big(\hat{\bar{a}}_w[k]+\xi\big)^2
\end{aligned}
$$

8 Observability: gates 1–3

$\big(S[k] := \sum_{j=k_0}^{k-1}\Delta\hat{\bar{w}}[j],\ S[k_0]=0;\ R_2$ the y_2 variance$\big)$

$$
\begin{aligned}
F_e[k]_{:,6} &= \Big[\,0,\ 0,\ 0,\ 0,\ \Delta\hat{\bar{w}}[k],\ 1\,\Big]^\top \\
H[k]_{:,6} &= \Big[\,0,\ 0\,\Big]^\top
\end{aligned}
$$

$$
\begin{aligned}
e_b[k] &= e_b[k_0] + S[k]\;e_{b_1}[k_0] \\
e_{\bar{a}_w}[k+1] &\supset \big(1-\hat{\bar{a}}_w\big)^2\Delta\hat{\bar{w}}[k]\,\Big(e_b[k_0] + S[k]\,e_{b_1}[k_0]\Big) \\
e_{y_2}[k] &\supset -\big(1-\hat{\bar{a}}_w\big)^2\nabla_d\bar{w}_d[k]\,\Big(e_b[k_0] + S[k]\,e_{b_1}[k_0]\Big)
\end{aligned}
$$

$$
\begin{aligned}
c[k] &= -\frac{\big(1-\hat{\bar{a}}_w[k]\big)^2 \nabla_d \bar{w}_d[k]}{\sqrt{R_2[k]}} \\
\mathbf{r}[k] &= c[k]\,\big[\,1,\ S[k]\,\big] \\
\det\begin{bmatrix}\mathbf{r}[k]\\ \mathbf{r}[k+1]\end{bmatrix} &= c[k]\,c[k+1]\,\Delta\hat{\bar{w}}[k] \\
G_2 &= \sum_k \mathbf{r}[k]^\top \mathbf{r}[k] \\
\det G_2 &= \Big(\sum_k c[k]^2\Big)\Big(\sum_k c[k]^2 S[k]^2\Big) - \Big(\sum_k c[k]^2 S[k]\Big)^2 = \Big(\sum_k c[k]^2\Big)^2 \operatorname{Var}_{c^2}(S) \\
\operatorname{Var}_{c^2}(S) &= \frac{\sum_k c[k]^2 S[k]^2}{\sum_k c[k]^2} - \Big(\frac{\sum_k c[k]^2 S[k]}{\sum_k c[k]^2}\Big)^2 \\
\sqrt{\big[G_2^{-1}\big]_{22}} &= \frac{1}{\sqrt{\sum_k c[k]^2\ \operatorname{Var}_{c^2}(S)}}
\end{aligned}
$$

$$
\det G_2 = 0 \iff c[k] = 0 \ \ \forall k \quad \text{or} \quad S[k] = \text{const on } \{k : c[k] \neq 0\}
$$

9 Production blocks

$\big(\mathbf{x} = [\,\delta\bar{w}_1,\ \delta\bar{w}_2,\ \delta\bar{w}_3,\ \bar{a}_w,\ b,\ b_1,\ \cdot,\ m_1,\ m_2\,]^\top$, slot 7 locked; $\mathbf{e}_i$ unit vector i; $w_T[k] = \bar{a}_w[k]\,\bar{f}_T[k]\big)$

$$
\begin{aligned}
\Delta\hat{\bar{w}}[k] &= \Delta\bar{w}_d[k] + (1-\lambda_c)\Big(\delta\hat{\bar{w}}_3[k] + \hat{m}_1[k] + \hat{m}_2[k] + y_1[k] - \delta\hat{\bar{w}}_1[k]\Big)\\
\mathbf{c}[k] &= \Big[-(1-\lambda_c),\ 0,\ (1-\lambda_c),\ F_{dw}[k],\ 0,\ 0,\ 0,\ (1-\lambda_c),\ (1-\lambda_c)\Big]^\top\\
e_{\Delta\bar{w}}[k] &= \mathbf{c}[k]^\top\mathbf{e}[k] + w_T[k]
\end{aligned}
$$

$$
\begin{aligned}
\bar{a}_w[k+1] &= \bar{a}_w[k] + b\big(1-\bar{a}_w\big)^2\Delta\bar{w}[k] + \tfrac{1}{2}\Big[b_1\big(1-\bar{a}_w\big)^2 - 2\,b^2\big(1-\bar{a}_w\big)^3\Big]\Delta\bar{w}[k]^2 + \mathcal{O}\big(\Delta\bar{w}^3\big)\\
\frac{1}{1-\bar{a}_w[k+1]} &= \frac{1}{1-\bar{a}_w[k]} + b[k]\,\Delta\bar{w}[k] + \tfrac{1}{2}\,b_1[k]\,\Delta\bar{w}[k]^2\\
b[k+1] &= b[k] + b_1[k]\,\Delta\bar{w}[k]
\end{aligned}
$$

$$
\begin{aligned}
e_{\bar{a}_w}[k+1] &= \big(\text{first order}\big) - 2\,\hat{b}\big(1-\hat{\bar{a}}_w\big)\Big(e_{\bar{a}_w}\,e_{\Delta\bar{w}} - \mathbb{E}\big[e_{\bar{a}_w}\,e_{\Delta\bar{w}}\big]\Big) + \big(1-\hat{\bar{a}}_w\big)^2\Big(e_b\,e_{\Delta\bar{w}} - \mathbb{E}\big[e_b\,e_{\Delta\bar{w}}\big]\Big)\\
&\quad + \tfrac{1}{2}\Big[\hat{b}_1\big(1-\hat{\bar{a}}_w\big)^2 - 2\,\hat{b}^2\big(1-\hat{\bar{a}}_w\big)^3\Big]\Big(e^2_{\Delta\bar{w}} - \mathbb{E}\big[e^2_{\Delta\bar{w}}\big]\Big) + \mathcal{O}(3)\\
e_b[k+1] &= \big(\text{first order}\big) + \Big(e_{b_1}\,e_{\Delta\bar{w}} - \mathbb{E}\big[e_{b_1}\,e_{\Delta\bar{w}}\big]\Big)
\end{aligned}
$$

$$
\begin{aligned}
\mathbb{E}\big[e_i\,e_{\Delta\bar{w}}\big] &= \big(P[k]\,\mathbf{c}[k]\big)_i\\
\mathbb{E}\big[e^2_{\Delta\bar{w}}\big] &= \mathbf{c}[k]^\top P[k]\,\mathbf{c}[k] + \kappa_T\,\hat{\bar{a}}_w[k]
\end{aligned}
$$

$$
\begin{aligned}
\delta\hat{\bar{w}}_1[k+1] &= \delta\hat{\bar{w}}_2[k]\\
\delta\hat{\bar{w}}_2[k+1] &= \delta\hat{\bar{w}}_3[k]\\
\delta\hat{\bar{w}}_3[k+1] &= \delta\hat{\bar{w}}_3[k] + \Delta\bar{w}_d[k] - \Delta\hat{\bar{w}}[k]\\
\hat{\bar{a}}_w[k+1] &= 1 - \frac{1}{\dfrac{1}{1-\hat{\bar{a}}_w[k]} + \hat{b}[k]\,\Delta\hat{\bar{w}}[k] + \tfrac{1}{2}\,\hat{b}_1[k]\,\Delta\hat{\bar{w}}[k]^2}\\
&\quad - 2\,\hat{b}\big(1-\hat{\bar{a}}_w\big)\big(P\mathbf{c}\big)_4 + \big(1-\hat{\bar{a}}_w\big)^2\big(P\mathbf{c}\big)_5\\
&\quad + \tfrac{1}{2}\Big[\hat{b}_1\big(1-\hat{\bar{a}}_w\big)^2 - 2\,\hat{b}^2\big(1-\hat{\bar{a}}_w\big)^3\Big]\Big(\mathbf{c}^\top P\,\mathbf{c} + \kappa_T\,\hat{\bar{a}}_w\Big)\\
\hat{b}[k+1] &= \hat{b}[k] + \hat{b}_1[k]\,\Delta\hat{\bar{w}}[k] + \big(P\mathbf{c}\big)_6\\
\hat{b}_1[k+1] &= \hat{b}_1[k]\\
\hat{m}_1[k+1] &= 0\\
\hat{m}_2[k+1] &= \hat{m}_1[k]
\end{aligned}
$$

$$
\begin{aligned}
F_e[k]_{3,:} &= \mathbf{e}_3^\top - \mathbf{c}^\top \\
F_e[k]_{4,:} &= \mathbf{e}_4^\top + \hat{b}\big(1-\hat{\bar{a}}_w\big)^2\mathbf{c}^\top - 2\,\hat{b}\big(1-\hat{\bar{a}}_w\big)\Delta\hat{\bar{w}}\;\mathbf{e}_4^\top + \big(1-\hat{\bar{a}}_w\big)^2\Delta\hat{\bar{w}}\;\mathbf{e}_5^\top \\
F_e[k]_{5,:} &= \mathbf{e}_5^\top + \hat{b}_1\,\mathbf{c}^\top + \Delta\hat{\bar{w}}\;\mathbf{e}_6^\top \\
F_e[k]_{6,:} &= \mathbf{e}_6^\top \\
F_e[k]_{8,:} &= \mathbf{0}^\top, \qquad F_e[k]_{9,:} = \mathbf{e}_8^\top \\
\mathbf{g}_T[k] &= \Big[\,0,\ 0,\ -1,\ \hat{b}\big(1-\hat{\bar{a}}_w\big)^2,\ \hat{b}_1,\ 0,\ 0,\ 1,\ 0\,\Big]^\top \\
Q[k] &= \kappa_T\,\hat{\bar{a}}_w[k]\;\mathbf{g}_T[k]\,\mathbf{g}_T[k]^\top \\
H_2[k]_6 &= 0
\end{aligned}
$$

Seed and prior

$\big(\bar{w}_0$ start height, w_c contact height, $\bar{a}_w(w_c)=0\big)$

$$
\begin{aligned}
1 &= \frac{1}{1-\bar{a}_w(\bar{w}_0)} + b(\bar{w}_0)\,(w_c-\bar{w}_0) + \tfrac{1}{2}\,b_1\,(w_c-\bar{w}_0)^2 \\
\hat{b}[0] &= \frac{1/\big(1-\hat{\bar{a}}_w[0]\big)-1}{\bar{w}_0-w_c} + \tfrac{1}{2}\,\hat{b}_1[0]\,(\bar{w}_0-w_c) \\
b(w_c) &= b(\bar{w}_0) - b_1\,(\bar{w}_0-w_c) \\
b\Big(\tfrac{1}{2}(\bar{w}_0+w_c)\Big) &= \frac{1/\big(1-\bar{a}_w(\bar{w}_0)\big)-1}{\bar{w}_0-w_c}
\end{aligned}
$$

$$
\begin{aligned}
e_b[0] &= e_b[0]\Big|_{e_{b_1}[0]=0} + \tfrac{1}{2}\,(\bar{w}_0-w_c)\,e_{b_1}[0] \\
\mathbb{E}\Big[e_{b_1}[0]\;e_b[0]\Big|_{e_{b_1}[0]=0}\Big] &= 0 \\
P_{56}[0] &= \tfrac{1}{2}\,(\bar{w}_0-w_c)\,P_{66}[0] \\
P_{55}[0] &= P_{55}[0]\Big|_{P_{66}[0]=0} + \tfrac{1}{4}\,(\bar{w}_0-w_c)^2\,P_{66}[0]
\end{aligned}
$$

$$
\begin{aligned}
e_b(w_c) &= e_b(\bar{w}_0) - (\bar{w}_0 - w_c)\, e_{b_1} \\
P_{55}(w_c) &= P_{55}[0] - 2\,(\bar{w}_0 - w_c)\, P_{56}[0] + (\bar{w}_0 - w_c)^2\, P_{66}[0] \\
P_{56}[0] = 0: \quad P_{55}(w_c) &= P_{55}[0]\Big|_{P_{66}[0]=0} + (\bar{w}_0 - w_c)^2\, P_{66}[0] \\
P_{56}[0] = \tfrac{1}{2}(\bar{w}_0 - w_c)P_{66}[0]: \quad P_{55}(w_c) &= P_{55}[0]\Big|_{P_{66}[0]=0} + \tfrac{1}{4}\,(\bar{w}_0 - w_c)^2\, P_{66}[0] \\
\sqrt{P_{55}(w_c)} &\leq \sup_{\text{env}} \Big|\, b_{\text{true}} - \tfrac{8}{9}\, \Big| \\
\Rightarrow\ P_{56}[0] = 0: \quad \sqrt{P_{66}[0]} &\leq \frac{\sqrt{\sup_{\text{env}} \Big|\, b_{\text{true}} - \tfrac{8}{9}\, \Big|^2 - P_{55}[0]\Big|_{P_{66}[0]=0}}}{\bar{w}_0 - w_c} \\
\Rightarrow\ P_{56}[0] = \tfrac{1}{2}(\bar{w}_0 - w_c)P_{66}[0]: \quad \sqrt{P_{66}[0]} &\leq \frac{2\sqrt{\sup_{\text{env}} \Big|\, b_{\text{true}} - \tfrac{8}{9}\, \Big|^2 - P_{55}[0]\Big|_{P_{66}[0]=0}}}{\bar{w}_0 - w_c} \\
\sqrt{P_{55}[0]}\Big|_{P_{66}[0]=0} = \sup_{\text{env}} \Big|\, b_{\text{true}} - \tfrac{8}{9}\, \Big| &\ \Rightarrow\ \sqrt{P_{66}[0]} \leq 0 \\
\sqrt{P_{66}[0]} &= \sup_{\text{env}} \big|\, b'_{\text{true}}\, \big|
\end{aligned}
$$